MARUTI SUZUKI

XCELerate 2026

Powering the Future of Mobility!









Platform Partner unstop

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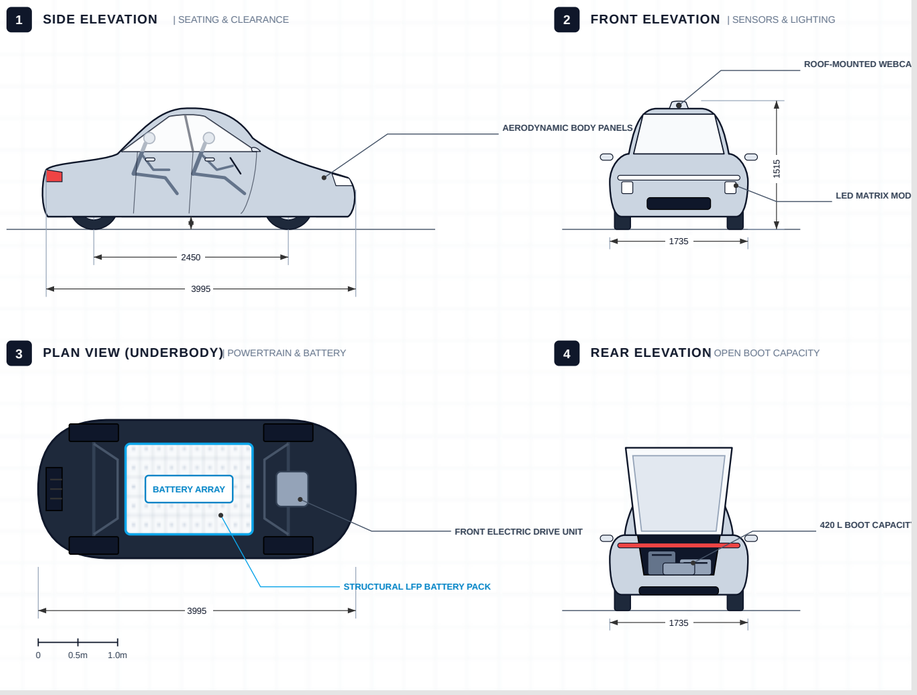
MACH1 — ELECTRIC MOBILITY 2035

Compact. Commercial. Electric. Connected.

XCELerate 2026

EV SEDAN CONCEPT - PACKAGING & UTILITY

4-VIEW ORTHOGRAPHIC LAYOUT



VEHICLE CONCEPT

BODY STYLE
Compact Sedan

Separate boot keeps passenger and luggage space distinct. Silhouette reads as professional — something riders respond to.

SIZE — 8-SEGMENT
3,995 mm · 2,450 mm WB

Ideal footprint for Tier 3 and Axle Tour S. Seats adults comfortably navigates city traffic, packs without drama.

SEATING
5-Seat, 2-Row · Fleet Build

Rear-biased layout. Flat floor (no tunnel), rear AC vents, durable leatherette. Boot: 400+ L beats Dire Tour S.

PRIMARY TARGET
Ride-share Fleet · Ola · Uber · BluSmart

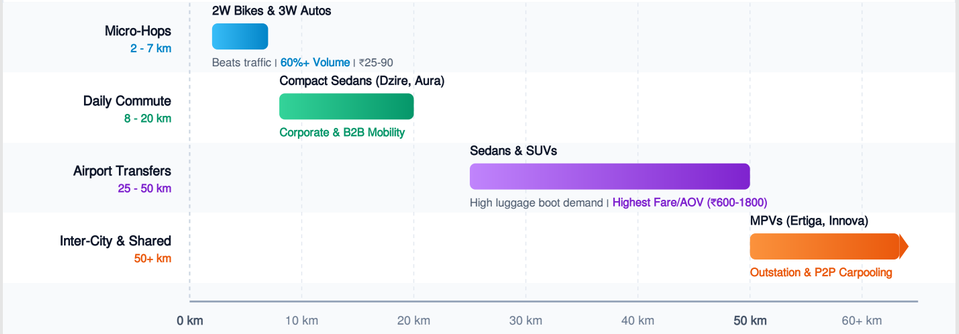
Dire Tour S proposed the formula. This takes the same commercial-ready brief and lifts it for operators already running EV fleets in metro cities.

SECONDARY TARGET
City Cab Aggregators

Clean vehicle mandates are pushing aggregators toward EV. Tiago EV is too small, nothing above it is fleet-priced. This fills that gap.

Mobility Trip Categories by Distance & Economics

Mapping vehicle formats, pricing, and volume drivers across urban transit radiuses.



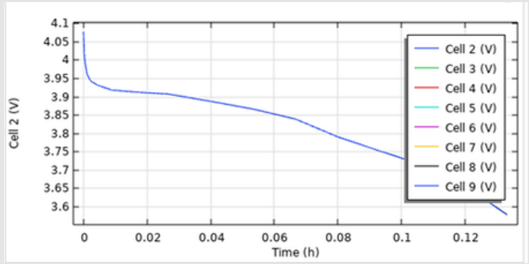
INDIA 2035 MARKET FORECAST

~$60B
Total Addressable Market
Mobility market scales from $13.75B (2026) at a 17.9% CAGR.

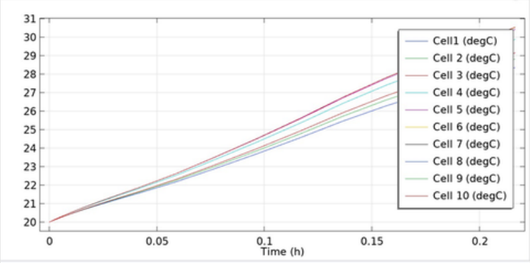
>$16B
Pure E-Hailing Volume
Online platforms capture majority share, driving commercial EV demand.

+19.1%
Target Segment CAGR
Airport transfers & Tier-2/3 expansion align perfectly with the sedan concept.

VOLTAGE DISCHARGE CHARACTERISTICS OF BATTERY CELLS



THERMAL PROFILE OF BATTERY CELLS DURING OPERATION



POWERTRAIN & ENERGY STRATEGY

PMSM Motor – 80–100 bhp, Water-Cooled

Single-speed reduction drive · BEV

Peak Efficiency	93–97%
Efficiency at 45°C	~2% drop
Energy Recovery	60–80% above 15 km/h
Rated Life	200,000+ km
Primary Wear Item	Bearings

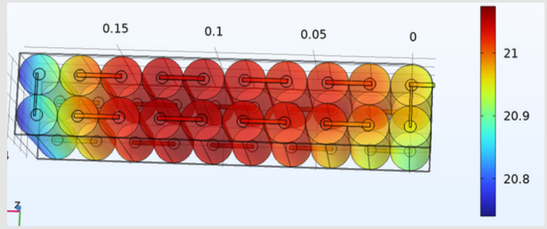
Battery Chemistry

	LFP	Na-ion
Energy density	90–205 Wh/kg	100–175 Wh/kg
Cycle life	2,500–9,000+	4,000–10,000+
Cost	₹8,500–12,000(kWh)	₹15,000–19,000(kWh)
Thermal runaway	~270° onset	Higher (Na-PO_4)
45°C sustained	20–30% life hit	Stable to 70–80°C
Peak charge	6C (Blade 2.0)	5C

DECISION

Lithium iron phosphate (LFP) battery is chosen as backup, with the main architecture designed around Na-Ion Battery. The car targets 2035 , We have enough runway for Na-Ion to close the cost gap and mature commercially. If it does, we drop it in. If it doesn't, LFP is the market default and the fallback is already proven.

THERMAL DISTRIBUTION MODEL OF LI-ION BATTERY PACK



TECHNICAL SPECIFICATIONS

POWERTRAIN	80–100 bhp PMSM Motor, Single-speed
BATTERY / RANGE	**45 kWh** LFP · **263 km** **200 km @ 75% DoD**
BRAKING	Front: Disc · Rear: Drum
WHEELS & TYRES	14" alloy · 175/65 R14 tubeless radial
GROUND CLEARANCE	**170 mm** unladen
LIGHTING	Full LED · DRL bar · Fog lamps
SAFETY	Airbags · ABS · **level 2 ADAS**
INTERIOR	5-seat · Ventilated seats · EPS · Panoramic sunroof (top variant)
BOOT SPACE	**400 L**
SEGMENT	Compact Sedan · B-segment

3,995	1,684	1,532	2,450
LENGTH MM	WIDTH MM	HEIGHT MM	WHEELBASE MM

APPROXIMATE TOTAL COST OF OWNERSHIP

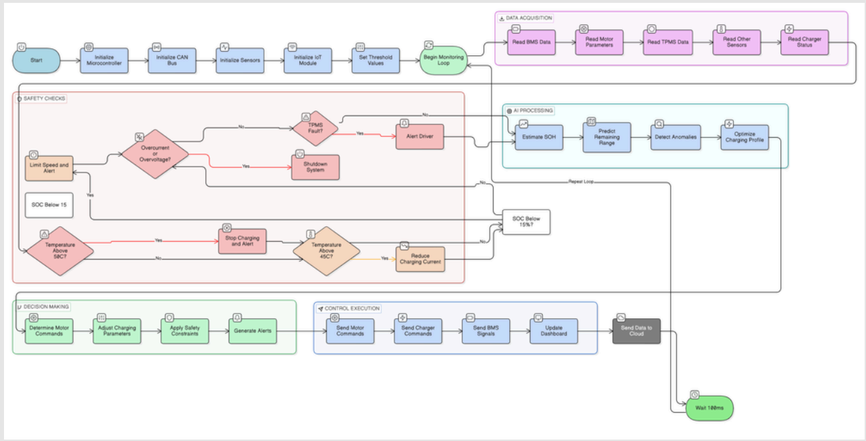
200 km/day - 26 days/month - ₹11/km fare - CNG ₹90/kg - Electricity ₹8/unit

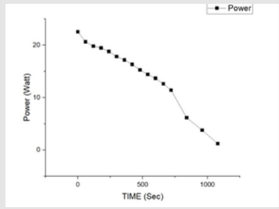
	EV (PROPOSED)	CNG
ON-ROAD PRICE	**₹14,00,000**	**₹8,51,465**
MONTHLY REVENUE	**₹57,200**	**₹57,200**
MONTHLY EMI	**₹29,740**	**₹18,090**
ENERGY COST / MONTH	**₹6,400**	**₹18,720**
MAINTENANCE / MONTH	**₹1,200**	**₹3,500**
INSURANCE / MONTH	**₹3,500**	**₹3,290**
NET TAKE-HOME (LOAN PERIOD)	**₹16,380**	**₹13,600**
5-YEAR TOTAL EARNINGS	**₹9.81 L**	**₹8.16 L**

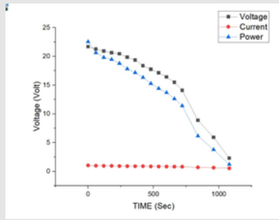
AFTER LOAN - YEAR 6+
DRIVER TAKE-HOME / MONTH **+₹14,400 / month advantage** **₹46,090** **CNG: ₹31,690**

①Intelligent EV Control System Architecture Flow

BMS | Motor/Inverter | GPS/NavIC | IMU | TPMS | Charging







② Collect:

- Battery SoC/SoH, cell temperature & voltage
- Motor/inverter temperature & fault data
- Tyre pressure and vehicle vibration
- Charging energy and charging history

3 CONNECT — India-Ready Data Layer

Vehicle → Edge Gateway → 4G/5G → Cloud

- MQTT/TLS for lightweight secure telemetry
- 72-hour local data buffer for network-shadow areas
- Critical safety events handled locally rather than depending on cloud connectivity

IMPLEMENTATION ROADMAP

1

2032–33
Development Lock-in
Na-Ion commercially available. Architecture finalised around Na-Ion primary cell. Reliance/Faradion supply agreement secured. LFP retained as base variant. Kharkhoda plant tooled for production.

2

2035
Launch
Delhi NCR, Mumbai, Bengaluru. 50,000–70,000 units/year. Na-Ion as primary offering, LFP as entry price point. Charging infrastructure via Reliance New Energy already in place.

3

2037+
Pan-India Scale
1,00,000+ units/year. Tier 2 expansion. LFP variant evaluated for phase-out depending on Na-Ion cost curve. Full domestic supply chain operational.

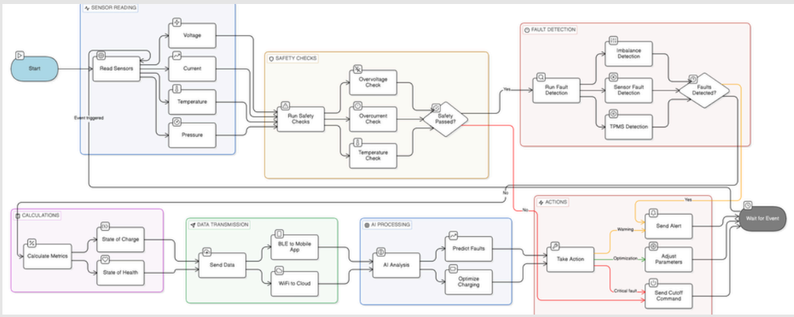
DIGITAL VALUE TARGETS

~15% ↓ Deadhead km
Smart dispatch
15–18% ↓ Charging cost
Smart charging
20–30% ↓ Unscheduled downtime
Predictive maintenance
≥7-day actionable maintenance warning
Detectable degradation
These are 2035 design targets

MARUTI SUZUKI VISION

★Experience: To make every journey an experience and every experience a memory, Maruti Suzuki puts people at the centre of everything it does.
★Technology: To improve driving experience and ensure a better future for the coming generations, Maruti Suzuki believes in green technology.
★Design: To give more than just a good-looking car, Maruti Suzuki ensure that every design element serves a purpose. More than design, it's design in technology.

③ AI-Based Battery Management and Vehicle Safety System



Battery Management
BMS data → SoH/degradation estimation → early service recommendation
Value: Protects battery life + reduces unexpected downtime.
Predictive Maintenance
Motor/inverter/thermal/vibration data → anomaly detection → workshop planning
Value: Less unscheduled downtime + better fleet availability.
Smart Fleet Dispatch
Demand + location + SoC + trip requirement → optimal vehicle allocation
Value: Lower deadhead kilometres + higher vehicle/driver utilisation.
Smart Charging
SoC + next-shift demand + tariff + charger availability → optimal charging schedule
Value: Lower charging cost + better depot utilisation.

THE INFRASTRUCTURE TRANSITION RISK

Maruti's spares and service network — 3,614 outlets, 480+ genuine parts distributors, growing at 19% YoY — is currently one of its most reliable revenue streams, generating roughly 10% of total company revenue and outpacing car sales growth. But this entire ecosystem is built for ICE. EV servicing is fundamentally different: fewer mechanical parts, high-voltage systems, battery diagnostics, OTA software. As Maruti transitions its fleet to EV, this recurring revenue model needs to be rebuilt in parallel — EV-certified technicians, battery health infrastructure, updated parts network. The transition window matters: whoever builds EV service density first owns the operator trust. For this car to sustain the same post-sale revenue logic, the service infrastructure needs to be EV-ready at launch, not retrofitted later.